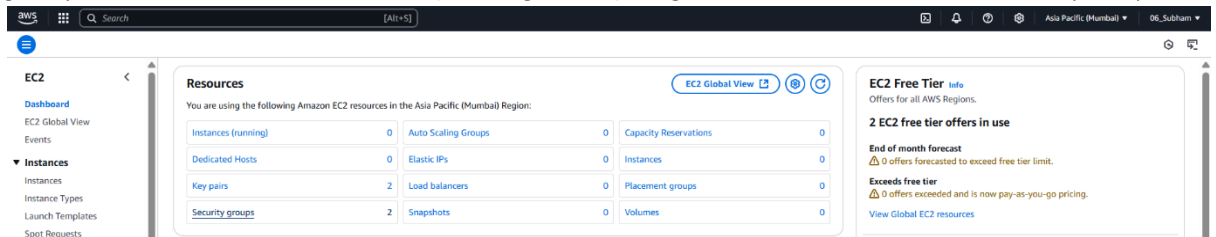


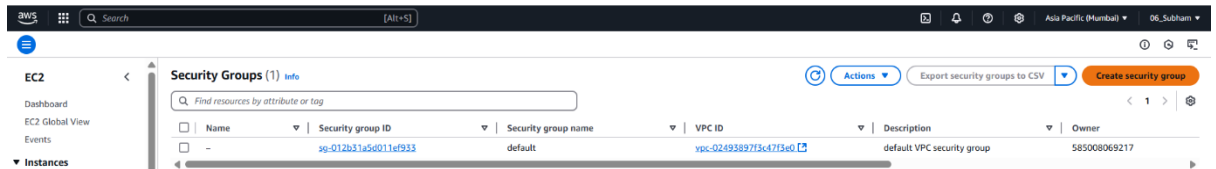
Assignment 10: Deploy a project from GitHub to EC2 by creating a new security group and user data.

Note: In case of a private GitHub repository we need to create a personal access token which is shown in Assignment 9 (Step 1 to Step 6)

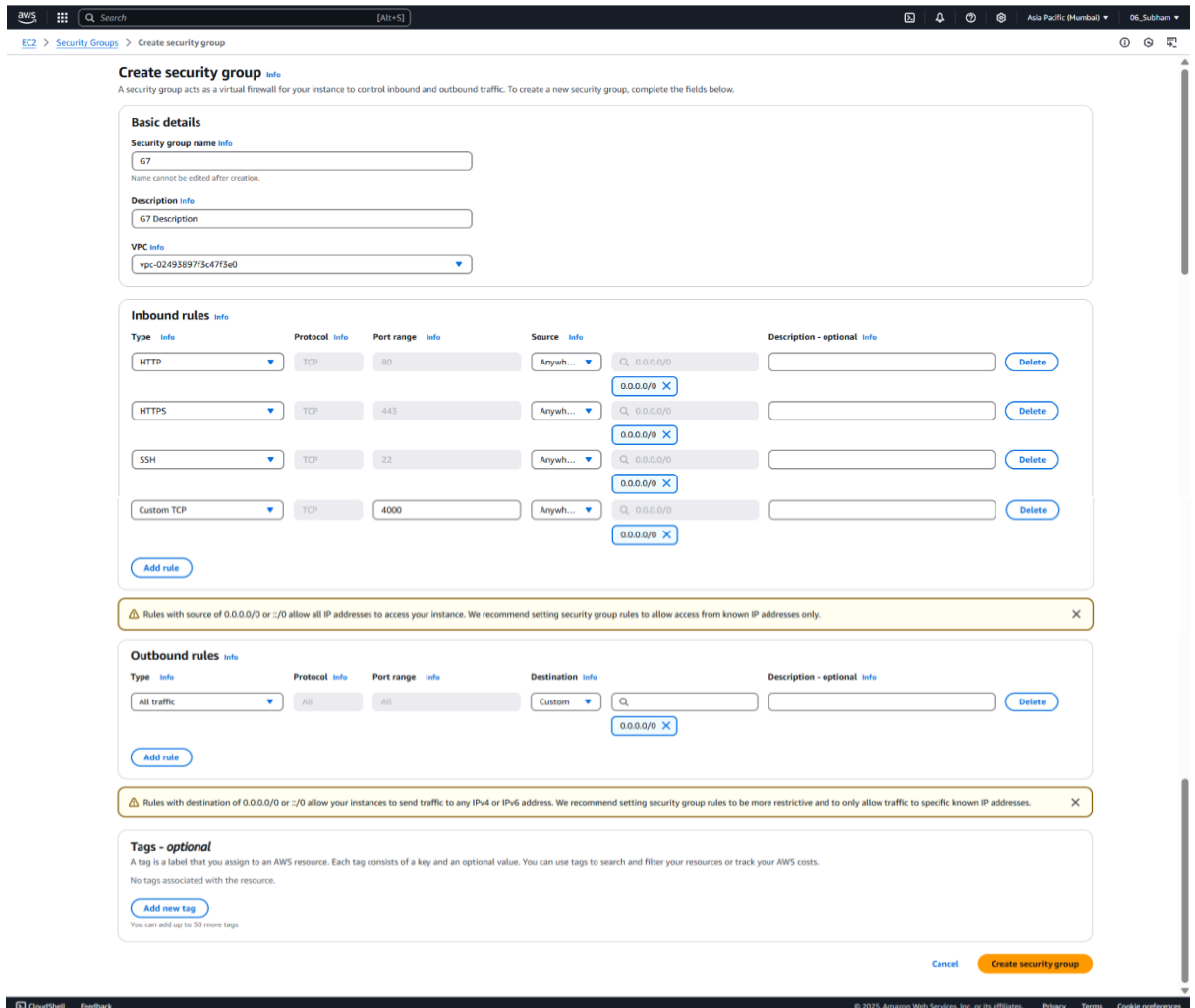
Step 1: Login to your AWS account and search for EC2 (Like Assignment 9) and go to the EC2 Dashboard. Click on Security Groups.



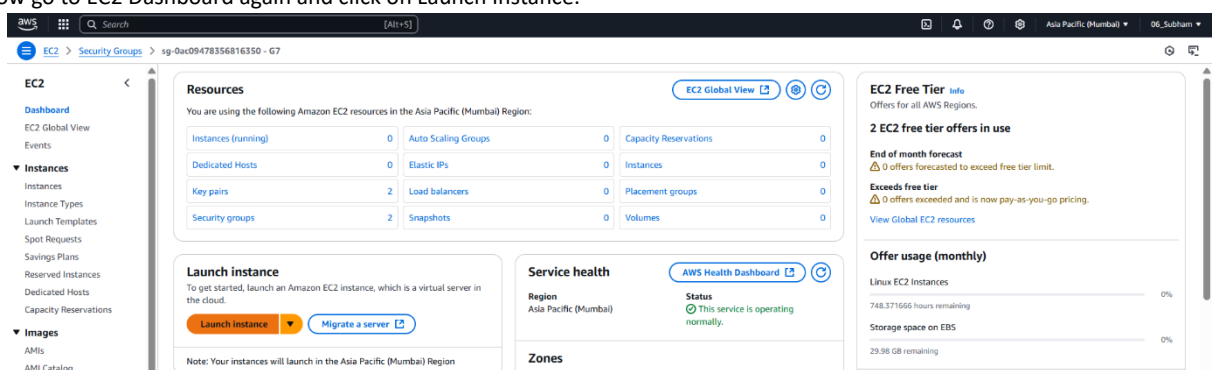
Step 2: Click on Create Security Group.



Step 3: Give a suitable name and description. Add the HTTP, HTTPS, SSH and Custom TCP Inbound Rules as follows and Click on Create Security Group.



Step 4: Now go to EC2 Dashboard again and click on Launch Instance.



Step 5: Give a Name of the Instance and select Ubuntu as the OS, select an existing key Pair and Existing Security Group (Which was created in the previous steps) and Under Advanced Settings and User Data write the commands as given in the image and then click on Launch Instance.

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name

G7_server

Add additional tags

Application and OS Images (Amazon Machine Image)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Free tier eligible

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type

ami-0e35ddab05955c5f57 (64-bit x86) / ami-0429b8da1c4d11a80 (64-bit ARMv8)

Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Ubuntu Server 24.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (http://www.ubuntu.com/cloud/services).

Canonical, Ubuntu, 24.04, amd64 noble image

Architecture

AMI ID

Publish Date

Username

Verified provider

64-bit (x86) ami-0e35ddab05955c5f57 2025-03-05 ubuntu

Instance type

Get advice

Free tier eligible

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Linux base pricing: 0.0124 USD per Hour

On-Demand Windows base pricing: 0.017 USD per Hour On-Demand RHEL base pricing: 0.0268 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0142 USD per Hour On-Demand SUSE base pricing: 0.0134 USD per Hour

Additional costs apply for AMIs with pre-installed software

All generations

Compare instance types

Key pair (login)

Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

keyG7

Create new key pair

Network settings

Info

Edit

Network

Info

vpc-02493897f3c47f3e0

Subnet

Info

No preference (Default subnet in any availability zone)

Auto-assign public IP

Info

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

Common security groups

Info

Select security groups

G7 sg-0ac09478356816350

VPC: vpc-02493897f3c47f3e0

Compare security group rules

Security groups that you add or remove here will be added to or removed from all your network interfaces.

Configure storage

Info

Advanced

1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

Add new volume

The selected AMI contains more instance store volumes than the instance allows. Only the first 0 instance store volumes from the AMI will be accessible from the instance

Click refresh to view backup information

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0x File systems

Edit

Advanced details

Info

Domain join directory

Info

Select

Create new directory

IAM instance profile

Info

Select

Create new IAM profile

Hostname type

Info

IP name

DNS Hostname

Info

Enable IP name (IPv4 (A record) DNS requests)

Enable resource-based IPv4 (A record) DNS requests

Enable resource-based IPv6 (AAAA record) DNS requests

Instance auto-recovery

Info

Select

Shutdown behavior

Info

Stop

Stop - Hibernate behavior

Info

Select

Termination protection

Info

Select

Stop protection

Info

Select

Detailed CloudWatch monitoring

Info

Select

Credit specification

Info

Standard

Placement group

Info

Select

Create new placement group

EBS-optimized instance

Info

Disable

Summary

Number of instances

Info

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64_read more

ami-0e35ddab05955c5f57

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

Cancel

Launch Instance

Preview code

Instance bandwidth configuration

Info

Select

Purchasing option

Info

☒ None

☐ Capacity Blocks

Launch instances for your active capacity blocks

☐ Spot Instances

Request Spot Instances at the Spot price, capped at the On-Demand price

RAM disk ID

Info

Select

Tenancy

Info

Select

RAM disk ID

Info

Select

Kernel ID

Info

Select

Nitro Enclave

Info

Select

Nitro Enclaves are not compatible with instance types that have less than 2 vCPUs.

License configurations

Info

Select

CPU options

Info

Configure CPUs for your instance to optimize performance and save on licensing costs.

☒ Use default CPU options

☐ Specify CPU options

The t2.micro instance type does not support configuring CPUs. To view instance types that support CPU options, see [Supported CPU options](#) in the EC2 User Guide.

Default active vCPUs

1

Total vCPUs

1

Metadata accessible

Info

Enabled

Metadata IPv6 endpoint

Info

Select

Metadata version

Info

V2 only (token required)

⚠️

For V2 requests, you must include a session token in all instance metadata requests. Applications or agents that use V1 for instance metadata access will break.

Metadata response hop limit

Info

2

Allow tags in metadata

Info

Select

User data - optional

Info

Upload a file with your user data or enter it in the field.

Choose file

```
#!/bin/bash
apt-get update -y
apt-get upgrade -y
apt-get install -y nginx
systemctl start nginx
systemctl enable nginx
apt-get install -y git
curl -sL https://deb.nodesource.com/setup_18.x | sudo -E bash -
apt-get install -y nodejs
```

☐ User data has already been base64 encoded

Note: Click on the check box of the instance in the dashboard section, the details of the instance is visible.

Step 7: From that click on 'Client key manager'. The following dialog box appears. Click on 'Import'.

Bitvise Client Key Manager - Cryptographic provider: Windows CNG (x86) with additions

Client Key Manager

You have the following SSH user authentication keys:

Location	Algorithm	Size	Passp...	SHA-256 Fingerprint	MDS Fingerprint	Bubble Bubble	Comment

Generate New
Modify
Remove

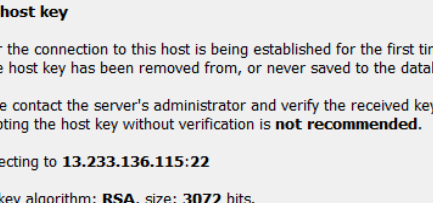
Import
Export
Change Passphrase
More

Step 8: Now select the file containing the client key and click on 'Open'.

File name: key7.pem Private Key Files (*.pkcs12;*.p12;*.p7b)

Open Cancel

Step 11: Now fill the details as follows and click on 'login'.



Host Key Verification

New host key

Either the connection to this host is being established for the first time or the host key has been removed from, or never saved to the database.

Please contact the server's administrator and verify the received key. Accepting the host key without verification is **not recommended**.

Connecting to **13.233.136.115:22**

Host key algorithm: **RSA**, size: **3072** bits.

SHA-256 Fingerprint:
/9HQ0wSrM4QAvjPy8DiC3piPC3iE2zxoXOMQWUSo

MD5 Fingerprint:
46:39:91:3f:00:9f:18:21:9d:54:f4:b2:71:a4:ac:d8

Bubble-Babble:
xifik-guros-bifyz-devom-remon-fyzer-semes-tyfom-gigok-zimek-naxax

Accept and Save Accept for This Connection Cancel

The screenshot displays the Bitvise SSH Server Control Panel interface. The 'Default profile' is selected, and the 'Server' tab is active. The configuration includes:

- Server:** Host (13.233.136.115), Port (empty), and an option to 'Enable obfuscation'.
- Authentication:** Username (ubuntu), Initial method (publickey), Client key (Global 1), and Elevation (Yes).
- Kerberos:** SPN (empty) and checkboxes for 'GSS/Kerberos key exchange', 'Request delegation', and 'gssapi-keyex authentication'.
- Proxy settings:** Empty section.
- Host key manager:** A list of host keys, including RSA and ECDSA keys, with their sizes and fingerprints.
- Client key manager:** Empty section.
- Log out:** A button at the bottom left.

```
ubuntu@13.233.136.115:22 - Bitvise xterm - ubuntu@ip-172-31-5-141: ~/DS
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

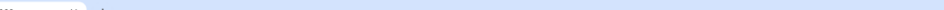
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-5-141:~$ git clone https://github.com/MirrorofMam-CS427/DS.git
```

```
ubuntu@13.233.136.115:22 - Bitwise xterm - ubuntu@ip-172-31-5-141: ~/DS
ubuntu@ip-172-31-5-141:~$ cd DS/
```

```
ubuntu@13.233.136.115:22 - Bitwise xterm - ubuntu@ip-172-31-5-141: ~/DS
ubuntu@ip-172-31-5-141:~/DS$ npm install
```

```
ubuntu@13.233.136.115:22 - Bitwise xterm - ubuntu@ip-172-31-5-141: ~/DS
ubuntu@ip-172-31-5-141:~/DS$ node index.js
Started server
```

A screenshot of a web browser window. The address bar shows a URL starting with '13.233.136.115:4000' and a warning icon with the text 'Not secure'. Below the address bar, the text 'Hello MCKVie' is visible.